

Lucas Zheng

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EDUCATION

McMaster University

Hamilton, Ontario

B.A.Sc. in Honours Computer Science Co-op; Minor in Statistics; GPA: 3.9/4.0

Sep 2024 – Apr 2028

Relevant coursework: Data Structures & Algorithms, Databases, Concurrent Systems, Computer Architecture, Applications of Machine Learning, Operating Systems, Software Engineering

EXPERIENCE

Software Developer (Part-Time)

Remote

Dishon Ltd.

Sep 2026 – Present

- Maintain and extend the production data-collection (PDC) system, resolving production issues, deploying new features, and supporting day-to-day operations.

Software Developer Intern

Vaughan, Ontario

Dishon Ltd.

May 2026 – Aug 2026

- Rebuilt the legacy production data-collection (PDC) system used by 200+ machine operators, cutting work-order lookup, sign-off, and clock-in/out response time from 15–20 seconds to under 1 second.
- Replaced manual recordkeeping with automated audit logging and daily reports, improving operational traceability and adhering to AS9100 and ISO 9001 compliance requirements.
- Maintained 99.97% production uptime over 90 days while managing Nginx, HTTPS/TLS, internal DNS, PM2, PostgreSQL migrations, and automated database backups.
- Built a TypeScript API layer integrating PDC, attendance, and administrative workflows with ABAS ERP using Next.js, Drizzle ORM, and PostgreSQL, with session-based authentication and role-based access control (RBAC).

Deli Associate

Burlington, Ontario

Fortinos Supermarket

Sep 2023 – May 2026

- Delivered efficient customer service and collaborated with team members in a fast-paced retail environment.

PROJECTS

TrailSense 🏠 | Node.js, Flask, TwelveLabs Pegasus/Marengo, Google Maps, OpenAI SDK

- Built a semantic video search engine for mountain-biking footage, enabling riders to retrieve relevant trails from natural-language prompts with timestamped results.
- Integrated TwelveLabs Marengo/Pegasus to generate scene-level embeddings, terrain summaries, and 0–10 difficulty ratings; extracted EXIF GPS/elevation metadata to associate footage with trail locations.

Toronto Airbnb Price Predictor 🏠 | Python, PyTorch, scikit-learn, Pandas, NumPy, Matplotlib

- Trained and evaluated a PyTorch regression model on 15,500+ Toronto Airbnb listings, achieving $R^2 = 0.46$ and \$56 MAE on held-out test data.
- Built a preprocessing and training pipeline with scikit-learn one-hot encoding and feature scaling, Adam optimization, L2 regularization, and validation-based early stopping.
- Visualized results with predicted-vs-actual scatterplots, loss curves, and feature-weight charts.

SyllaBuddy 🏠 | Python, TypeScript, Flask, Next.js, Tailwind CSS, OpenAI SDK

- Built a full-stack syllabus analyzer using Next.js, Flask, and OpenAI SDK to extract and summarize course information from uploaded PDFs.
- Developed a Flask API with pdfplumber to parse unstructured syllabi into structured course data, including assessments, deadlines, and textbook information.
- Designed a responsive React and Tailwind CSS dashboard for visualizing parsed course data and automating manual syllabus review and textbook lookup.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL

Frameworks & Libraries: Next.js, React, Flask, Tailwind CSS, scikit-learn, Matplotlib, Pandas, PyTorch, NumPy

Tools & Infrastructure: PostgreSQL, Drizzle ORM, Git, Nginx, PM2, ABAS ERP